

Illustration 4

Compare portfolio performance using Sharpe's and Treynor's measures for the following portfolios:

	Average Return (%)	Standard Deviation	Beta
Portfolio A	14%	0.25	1.25
Portfolio B	10%	0.15	1.10
Market Index	12%	0.25	1

The risk-free rate of return is 8%.

Solution:

$$\text{Sharpe's Measure} = \frac{R_p - R_F}{\sigma}$$

$$\text{Portfolio A} = \frac{14 - 8}{0.25} = 24\%$$

Rank

..... (1)

$$\text{Portfolio B} = \frac{10 - 8}{0.15} = 13.33\%$$

..... (2)

$$\text{Market Index} = \frac{12 - 8}{0.25} = 16\%$$

(3)

Portfolio A has out performed the market and Portfolio B has under performed the market.

$$\text{Treynor's Measure} = \frac{R_p - R_F}{\beta}$$

$$\text{Portfolio A} = \frac{14 - 8}{1.25} = 4.8\% \quad \text{Rank} \quad \dots (1)$$

$$\text{Portfolio B} = \frac{10 - 8}{1.10} = 1.82\% \quad \dots (2)$$

$$\text{Market Index} = \frac{12 - 8}{1} = 4\% \quad \dots (3)$$

As per Treynor's Measure, Portfolio A is better than Portfolio B and market index.

Illustration 5

Following information is given in respect of three Mutual Funds and Market:

Mutual Funds	Average Return	Std. Deviation	Beta
A	12	18	1.1
B	10	15	0.9
C	13	20	1.2
Market	11	17	1.0

The mean risk-free rate 6%. Calculate the Treynor's measure and Sharpe's measure and rank the portfolios.

Solution:

$$\text{Sharpe's Measure} = \frac{R_P - R_F}{\sigma}$$

$$\text{Mutual Fund A} = \frac{12 - 6}{0.18} = 33.33\% \quad \dots (2)$$

$$\text{Mutual Fund B} = \frac{10 - 6}{0.15} = 26.67 \quad \dots (4)$$

$$\text{Mutual Fund C} = \frac{13 - 6}{0.20} = 35\% \quad \dots (1)$$

$$\text{Market Index} = \frac{11 - 6}{0.17} = 29.41\% \quad \dots (1)$$

As per Sharpe's Measure, the Mutual Fund C and Mutual Fund A has outperformed the market and Mutual Fund B has underperformed the market.

$$\text{Treynor's Measure} = \frac{R - R_F}{\beta}$$

$$\text{Mutual Fund A} = \frac{12 - 6}{1.1} = 5.45\%$$

Rank

.... (2)

$$\text{Mutual Fund B} = \frac{10 - 6}{0.9} = 4.44\%$$

.... (4)

$$\text{Mutual Fund C} = \frac{13 - 6}{1.2} = 5.83\%$$

.... (1)

$$\text{Market Index} = \frac{11 - 6}{1.0} = 5\%$$

.... (3)

As per Treynor's Measure, Mutual Fund C is better than A, B and Market Index.

Illustration 6

The actual results of the portfolios and market index during the past three years are given below:

Portfolio	Return	Beta	Risk Free Rate
Birla	15%	1.2	9
Kotak	16%	1.5	9
Reliance	12%	0.8	9
Market index	13%	1.0	9

You are required to rank as per Jensen's measure of portfolio return.

Solution:

Jensen's Measure:

$$R = R_F + \beta (R_M - R_F)$$

Where,

R = Rate of Return

R_F = Risk-free rate

β = Beta

R_M = Rate of Market Portfolio

R_F = Risk-free rate

Jensen's Measures

$$R = R_F + \beta (R_M - R_F)$$

$$\begin{aligned} \text{Birla} &= 9 + 1.2 (13 - 9) \\ &= 9 + 4.8 \\ &= 13.8 \end{aligned}$$

$$\begin{aligned}\text{Kotak} &= 9 + 1.5 (13 - 9) \\ &= 9 + 6 \\ &= 15\end{aligned}$$

$$\begin{aligned}\text{Reliance} &= 9 + 0.8 (13 - 9) \\ &= 9 + 3.2 \\ &= 12.2\end{aligned}$$

$$\begin{aligned}\text{Market Index} &= 9 + 1.0 (13 - 9) \\ &= 9 + 4 \\ &= 13\end{aligned}$$

Now, Jensen's Measure = $R_p - R$

	Rank
Birla = $15 - 13.8 = 1.2\%$	 (1)
Kotak = $16 - 15 = 1\%$	 (2)
Reliance = $12 - 12.2 = 0.2\%$	 (3)
Market = $13 - 13 = 0$ (by definition)	 (4)

As per Jensen's Measure, Portfolio performance of Birla is Rank 1, Kotak is Rank 2, Reliance is Rank 3.

Illustration 7

Three mutual funds have reported the following return and risk for last five years.

Growth rate	RETURN	Standard Deviation	Beta
HDFC	15%	15%	1.10
ICICI	13%	16%	1.25
UTI	12%	10%	0.90

Compare the portfolio performance of Mutual Fund as well as market using Sharpe's and Treynor's Index. Which portfolio has better performance?

Solution:

$$\text{Sharpe's Measure} = \frac{R_p - R_F}{\sigma}$$

	Rank
HDFC = $\frac{15 - 10}{0.15} = 33.33\%$	 (1)
ICICI = $\frac{13 - 10}{0.16} = 18.75\%$	 (3)

$$\text{UTI} = \frac{12 - 10}{0.10} = 20\% \quad \dots (2)$$

As per Sharpe's Measure, HDFC is better than UTI & ICICI.

$$\text{Treynor's Measure} = \frac{R - R_F}{\beta}$$

	Rank
$\text{HDFC} = \frac{15 - 10}{1.10} = 4.55\%$	 (1)

$\text{ICICI} = \frac{13 - 10}{1.25} = 2.4\%$	 (2)
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$\text{UTI} = \frac{12 - 10}{0.90} = 2.22\%$	 (3)
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As per Treynor's measure, HDFC is better than ICICI and UTI.